

Understanding Speed-Dating Matches Using Autoencoder Derived

Latent Features and Mixed-Effects Modeling

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1. Introduction

The system and sanctity of mate selection, and the broader institution of marriage, is one that has held strong throughout much of human history. Choosing a mate whose features best compliment yours, and with whom you can continue your family history through, is and always will be a high priority compared to other factors such as location or timing (Zimmerman, 1950). Complimentary features, rather than social and religious expectations, is a relatively new idea of mate selection (Zimmerman, 1950). Personal choices are rising to high importance when selecting a partner for life, although research has had trouble pinning down just what characteristics drive mate selection (Fisman et al., 2006). Mate preferences have been observed to follow gendered differences (Mroz, 1987). Women have been expected to be the foundation of the family, taking care of child-rearing, attending to their husband, doing all household chores, etc. Men have been expected to work a career outside of the house, being the main source of financial support and upholding the integrity of home (Mroz, 1987; Fisman et al., 2006). These observed differences help build a case as to what features drive people to choose a certain other for life.

Speed-dating has become a popular method of choosing a mate fueled by one's own personality and life experiences. Speed-dating environments, while not exact experiments definitionally, provide an observational environment that can be equitable to an experiment. Speed-dating randomly partners people in such a way that, at some point, every unit has been "treated" multiple times. This is as though it was a panel-study where "treatment" would be exposure to a partner and the outcome of interest is a binary "match" (did two subjects mutually "match" with each other after exposure to each other's personalities and lives?) (Gerber & Green, 2012). Trials, in this case periods in which a subject is exposed to a partner, are conducted multiple times per-subject, and each trial allows for the subject to determine what individual-level preferences and features they liked/disliked about their partner pre- and post-exposure.

This speed-dating design, as seen in Fisman et al. (2006), allows for monitoring of possible differences and strengths of explicit and implicit features driving the matching process for subjects. Using this, we aim to investigate and rigorously test multiple models of analysis to better understand the most potent features of partner selection. This paper uses a two stage modeling framework combining unsupervised representation learning with hierarchical statistical modeling. First, an autoencoder is trained to compress a large set of participant characteristics, preference profiles, self ratings, partner ratings, and behavioral variables into a small set of latent dimensions. These latent representations provide a lower dimensional summary of the complex information contained within the original survey represented. Second, the learned latent features are incorporated into mixed-effects statistical models that account for the repeated measures structure of the speed dating experiment, where multiple dating interactions are observed for each participant. These approaches lead to the following research question:

***RQ1:** Can a small set of latent features, extracted by an autoencoder from demographic characteristics, self-rating, partner evaluations, preference profiles, and personal interests, predict the probability of a mutual match in a speed dating environment while properly accounting for repeated observations within participants?*

We hypothesize that the latent features learned by the autoencoder will collectively exhibit significant associations with the probability of a mutual match. More precisely, the autoencoder compresses 63 observed variables into an 8 dimensional latent representation through a bottleneck encoder-decoder architecture, which forces the model to learn compact summaries of participant preferences, evaluations, and behavioral characteristics. These latent dimensions are expected to capture underlying patterns relevant to romantic attraction and matching outcomes that may not be apparent from individual survey variables alone.

The code for this project can be found here: <https://github.com/arielktran/SpeedDating>

2. Methods

2.1 Data Processing

2.1.1 Cleaning and Preparing Dataset

The speed dating dataset comes from “Gender Differences in Mate Selection: Evidence From a Speed Dating Experiment” (Fisman et al., 2006). This experiment was designed to investigate individual level decision making in relation to various attributes and personal demographic details. Participants were recruited through fliers and emails from Columbia University. Speed dating events were hosted at a popular bar nearby campus. There were a total of 20 waves, grouped up speed dating participants, across 13 different days. There were 1-2 waves per experiment day. Each wave had a range of 12-44 people per wave. Female participants were seated throughout the entirety of the event, and males would rotate through. Participants had 4 minutes to converse and 1 minute to fill out their score cards. These score cards collected information on partner attribute rating, the likelihood of the other person saying ‘yes’ for the participant, and if both participants have met each other before. Matches occurred when both participants recorded their decision as ‘yes’ on each other.

Demographic details were collected during the pre-survey that was filled out when first registering to this study. During the event, partner ratings and decisions were recorded on the score cards. The day after, post-event surveys Subjects must complete the post-event survey in order to be sent their potential matches and match contact details. Of those who completed the post-event survey and received matches, a final follow up survey was sent 3-4 weeks after the participant’s speed dating event. The pre-survey, post-survey, and final follow-up survey had participants repeatedly answer attribute important and rating questions. These questions asked respondents to distribute 100 points across 6 attributes: attractiveness, sincerity, intelligence, fun, ambition, and having shared interests/hobbies. A simplified overview of the repeated questions

are listed below. The repeated attribute questions document explicit preferences and potentially different implicit preferences as participants progress through the speed dating cycle.

Attributes Importances and Preferences (asked Pre and Post Speed Dating)

- What attributes do you look for in the opposite sex?
- What attributes do you think your fellow males/females look for on a date?
- What attributes do you think the opposite sex looks for on a date?

Attribute Ratings (asked Pre and Post Speed Dating)

- Rate your own attributes.
- How do you think others would rate your attributes?

One-off Attribute Questions

- During speed dating - Rate your partner's attributes.
- Post-speed dating - Think back to your speed dating decisions, re-evaluate the importance of attributes in your decisions.

Missing data was a notable problem here, due to the inherent nature of surveys. Because of the short one-minute time frame for surveys during the speed dating event, it is likely that some participants rushed through their answers and left some questions unanswered. Additionally, the post-event survey being required for participants to be sent their matches means that the post response rate hinges on the belief that potential matches were made in the first place.

To evaluate the missing data rates, the proportion of missing answers for every survey question and data column was visualized in a histogram (see figure 1). This gave insight into how to reduce features to a manageable amount, starting with dropping irrelevant columns to the current investigation as over 46.15% of the columns had 20% or more of their data missing. All columns including the suffix (x_s) are columns asking participants to rate themselves and a potential, ideal partner on certain attributes halfway through the night of the speed-dating event. These columns, while interesting to note, temporally had no relevance to the investigation alongside having heavy data missing due to researchers only asking participants for some, but not all, of the waves.

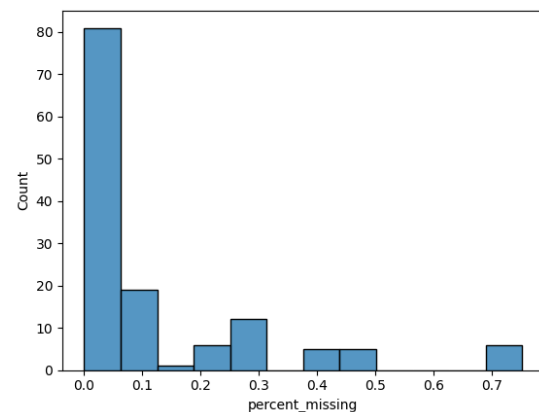
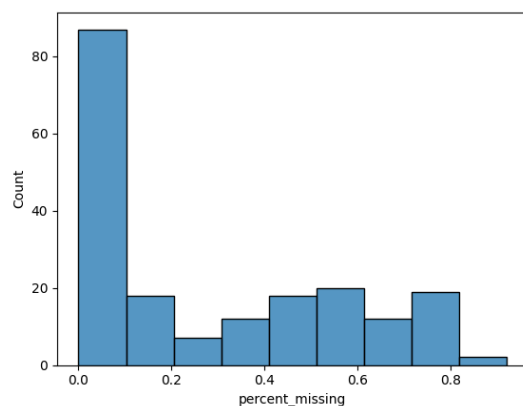


Figure 1. A histogram demonstrating percent of data missing per column pre-cleaning.

Figure 2. A histogram demonstrating percent of data missing per column post-cleaning.

Additionally, all responses from the final follow up survey had the lowest response rates because they were only sent to participants who had matches and finished the post-survey. The conditional logic of each subsequent survey leads to smaller subsets in addition to inherent survey attrition. Furthermore, attribute columns of little relevance to participant demographics were dropped, such as ‘undergraduate degree (undergra)’ as most participants were undergraduates with no degree yet. Finally, columns relating to non-attributes were dropped. After these changes, column data missingness was again re-evaluated. The filtering of the columns led to less missing data overall, with only 25.19% of columns having 20% or more of their data missing (see figure 2). The final step to cleaning the data was converting it from wide to long format. Columns for attributes were collapsed from ‘attribute_timePeriod’ to ‘attribute’ (such as attr_1, attr_2, attr_3 → attr). In this format, every row now represents each date from a participant’s perspective at one point in time. Repeated observations are seen per participant to demonstrate each time period and partner they had.

2.1.2 Exploratory Data Analysis

The Speed Dating dataset was first examined to understand its dimensionality, hierarchical structure, missingness patterns, feature scales, and suitability for both autoencoder-based representation learning and mixed-effects statistical modeling. The transformed long-format dataset contained 37,890 observations and 75 variables, consisting of demographic characteristics, dating preferences, partner evaluations, behavioral variables, and matching outcomes. The binary outcome variable, *match*, indicated whether mutual interest occurred following a speed-dating interaction.

The dataset exhibited a hierarchical repeated measures structure. Each participant (*iid*) was observed across multiple speed dating interactions with different partners. With the total observations, there were 362 unique subject partners, 503 unique partners, across 20 waves. Participants contributed between 10 - 231 observations.

Measure	Observations per Subject	Observations per Partners
Minimum	10	5
Maximum	231	187
Mean	104.67	75.33
Median	100.00	70.00

Table 1. Summary of observation counts by subject (*iid*) and partner (*pid*).

It is worth noting that this repeated measures design violates the independence assumption required by ordinary logistic regression as observations from the same participants

are correlated. Also, the maximum waves attended per participants and partners is 1, therefore, participants are nested within waves rather than crossing multiple events. This observation justified the use of participant level random effects rather than crossed random-effects structures.

The binary outcome variable, *match*, was also examined. Approximately 17.2% of observations resulted in a successful match, while the remaining 82.8% resulted in no match. This class imbalance suggested that evaluation metrics beyond simple accuracy would be necessary when assessing predictive performances.

Patterns of missing data were investigated across all variables. Missingness was generally low after the cleaning process, with the exception of a variable representing the subjects' shared interests & hobbies, which contained approximately 36.95% missing observations. Because of the substantial amount of missing information, this variable *shar* and its associated partner-rating variables were removed from further analyses. Remaining variables contained negligible levels of missingness and were kept.

The cardinality of categorical variables was also evaluated. Several variables, including field of study, hometown, zip code, and career aspirations, contained hundreds of unique categories. Inclusion of these variables would dramatically increase model complexity while contributing relatively sparse information. Therefore, these high-cardinality variables were excluded from the modeling pipeline. In contrast, low-cardinality categorical features such as *pre_post* and *rating_ref_group* were retained and later transformed using one-hot encoding.

Finally, the distribution and variance of numerical variables were examined. Several preference allocation variables exhibited larger variances than other features which indicates that feature scaling would be needed prior to training the autoencoder. Standardization was therefore applied during preprocessing to ensure that no individual variable dominated the reconstruction objective solely due to its numerical scale.

2.1.3 Further Pre-Processing

Following EDA, additional preprocessing steps were performed to prepare the data for representation learning and mixed-effects modeling. All the preprocessing operations were conducted after partitioning the data into training, validation, and test sets split by subjects to prevent information leakage from influencing model performance estimates. Following this split, administrative structural identifiers (*iid*, *pid*, *partner*, *wave*, *round*, *position*, *order*), the leakage inducing decision variables (*dec*, *dec_o*), the high-cardinality string features (*field*, *from*, *zipcode*, *career*), and both shared interest features (*shar*, *shar_o*) were removed to eliminate trivial predictors and reduce dimensionality. The remaining categorical variables were then transformed using dummy encoding with reference category coding (k-1 selection) to avoid multicollinearity before being combined with the numerical features. Next, minor missingness was addressed by imputing missing values using median values calculated exclusively from the training set. To ensure features on disparate numerical scales did not dominate the autoencoder's reconstruction objective, all predictors were standardized using the training set mean and standard deviation. These imputation and standardization parameters were applied unchanged to

the validation and test sets to maintain strict data separation. These steps produced a final modeling dataset of 63 standardized predictor variables which serves as input to the representation learning stage.

2.2 Autoencoder Representation Learning

To reduce dimensionality while preserving the information contained within the original predictors, an autoencoder neural network was trained using the preprocessed training data. Autoencoders are unsupervised learning models that learn compressed latent representations by reconstructing their inputs after passing them through a lower dimensional bottleneck layer. The network consists of an encoder and decoder architecture. The encoder compresses the 63 dimensional input space through two hidden layers before projecting observations into an 8 dimensional latent space. The encoder followed the architecture:

$$63 \rightarrow 48 \rightarrow 24 \rightarrow 8$$

The decoder mirrored this structure to reconstruct the original feature vector:

$$8 \rightarrow 24 \rightarrow 48 \rightarrow 63$$

Each hidden layer used Rectified Linear Unit (ReLU) activation functions. Batch normalization layers were included after each linear transformation to stabilize training and reduce sensitivity to initialization. Dropout regularization ($p=0.2$) was applied throughout the network to reduce overfitting and encourage more robust latent representations. The network was trained using the Adam optimizer with a learning rate of 0.001 and a batch size of 256 observations. Reconstruction performance was evaluated using Mean Squared Error (MSE) loss. Early stopping was implemented using validation set reconstruction loss with a patience threshold of 15 epochs. Training terminated when validation performance stopped improving, and the model parameters corresponding to the lowest validation loss were retained. After training, the encoder component was used to transform every observation into an 8 dimensional latent representation. These latent variables, denoted Z1 through Z8, served as the predictors in the downstream mixed-effects models.

2.3 Mixed-Effects Modeling

The latent variables generated by the autoencoder were incorporated into two mixed-effects modeling frameworks designed to account for the repeated-measures structure of the speed-dating experiment. Because each participant contributed multiple observations, standard regression would violate the assumption of independent observations. To account for within-participant correlation, both models incorporated participant specific random intercepts. The model included both main effects and gender by latent feature interaction terms with the following formula as both:

$$\text{match} \sim Z1 + Z2 + \dots + Z8 + \text{gender} + \text{gender} : Z1 + \dots + \text{gender} : Z8 + (1 \mid \text{iid})$$

The first model was a Mixed-Effects Linear Probability Model (MELPM) and was fitted using statsmodels MixedLM with grouping variable iid. This adds a random intercept for each

participant, accounting for the baseline “matchability” each subject brings independent of any particular partner’s features. The linear probability model outputs predictions on the real line, which must be clipped to $[0, 1]$ for evaluation purposes. This clipping is a known limitation of LPMs applied to binary outcomes, as predictions outside the unit interval are theoretically incoherent and produce artificially inflated log-loss when severely clipped values are passed to logarithmic scoring rules. The model’s information criteria (AIC, BIC) returned NaN values, indicating numerical instability in the log-likelihood computation likely attributable to these out-of-bounds fitted values. To address this caveat of our baseline model given the binary outcomes, we implemented a second model.

Our second model was estimated using a Bayesian Mixed-Effects Logistic Regression (MELoGR). This model used the same fixed-effect specification but constrained predicted probabilities to the interval $[0,1]$ and estimated participant-level random effects through a Bayesian variance-component framework.

Both models were trained using only the training partition and evaluated on previously unseen participants contained within the testing partition.

2.4 Model Evaluation

Model performance was evaluated using multiple complementary metrics. Because successful matches represented only approximately 17.2% of observations, accuracy alone was considered insufficient for assessing predictive quality. Discriminative performance was measured using the Area Under the Receiver Operating Characteristic Curve (AUC-ROC), which evaluates a model’s ability to correctly rank positive observations above negative observations across all possible classification thresholds. Classification accuracy was also reported using a probability threshold of 0.50 to facilitate comparison with a simple majority-class baseline.

Probability calibration was evaluated using Log Loss and Brier Score. Log Loss penalizes overconfident incorrect predictions and rewards well-calibrated probabilities, while the Brier Score measures the mean squared difference between predicted probabilities and observed outcomes. Confusion matrices, calibration curves, and ROC curves were generated to visualize predictive performance and model behavior. Autoencoder performance was assessed separately using reconstruction Mean Squared Error (MSE) on the validation and testing partitions. Training and validation loss curves were monitored throughout optimization to assess convergence and potential overfitting.

3. Results

3.1 Autoencoder Training and Representation Learning

The autoencoder converged successfully during training and demonstrated stable learning behavior throughout optimization. **Figure 3** displays the training and validation loss trajectories over the course of optimization. The training loss decreased monotonically from 0.991 to 0.730, while the validation loss improved from 0.873 to a minimum of 0.725 at epoch 52. Early

stopping was triggered after 67 epochs when validation performance ceased improving for 15 consecutive epochs, indicating that additional training was unlikely to improve generalization performance. The close correspondence between training and validation loss curves suggests that the combination of dropout regularization and batch normalization successfully limited overfitting despite the relatively large number of observations. The final validation reconstruction loss was 0.725, while the held-out test reconstruction mean squared error (MSE) was 0.703, indicating that the learned latent representation generalized well to previously unseen participants.

Further examination of reconstruction error revealed minimal differences between matched and unmatched observations. Non-matches exhibited a mean reconstruction error of 0.702, whereas matches exhibited a mean reconstruction error of 0.707. This finding suggests that the autoencoder learned a general low-dimensional representation of participant characteristics rather than encoding information specific only to matching outcomes. These latent dimensions were subsequently used as predictors in the mixed-effects models.

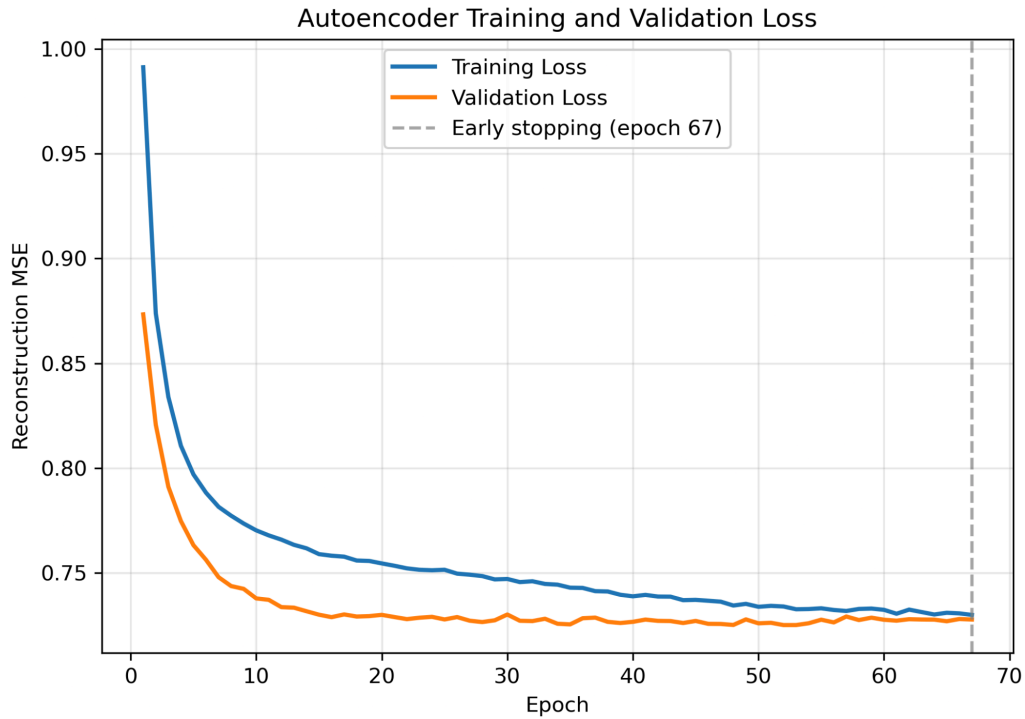


Figure 3: A line plot showing the training and validation reconstruction MSE across epochs until early stopping.

3.2 Interpretation of Learned Latent Dimensions

Because autoencoder latent variables are not directly interpretable, post-hoc correlations were computed between the latent dimensions and the original survey features. The strongest correlations for each latent dimension provide insight into the types of information captured by the learned representations. **Figure 4** presents the full correlation structure between the eight learned latent dimensions and the original 63 predictor variables. The heatmap corroborates the

post-hoc interpretations summarized in **Table 2**. For example, Z1 loads heavily on self-valuation and intelligence, Z4 separates cultural interests (museums, theater, art) from athletics (sports), and Z8 captures music and social enthusiasm. The clear, distinct correlation patterns confirm that the autoencoder compressed the feature space into interpretable, orthogonal themes relevant to mate selection.

Latent Dim.	Tor Correlates (r)	Interpretation
Z1	self_value(+0.67), intel(+0.49), sinc(+0.38)	self-perceived personal quality and positive self-evaluation.
Z2	music(+0.43), fun_o(+0.42), like_o(+0.41)	immediate partner attraction and perceived romantic compatibility
Z3	fun_o(+0.54), attr_o(+0.53), like_o(+0.53), gaming(-0.46)	favorable evaluations of dating partners vs. gaming interest
Z4	museums(+0.58), theater(+0.57), art(+0.56), sports(-0.5)	cultural versus athletic preference spectrum
Z5	date(+0.42), race(+0.39), go_out(+0.35), age(-0.33)	broader social and demographic characteristics
Z6	self_value(-0.7), intel(-0.52), imprace(+0.37)	low self-valuation
Z7	self_value(-0.56), intel(-0.44), like_o(-0.44)	low partner liking and low self-valuation
Z8	concerts(+0.49), music(+0.48), exphappy(+0.40)	social optimism and recreational engagement

Table 2: Top Correlates of each Autoencoder Latent Dimension with Original Features
(Pearson r on Training Set)

Collectively, these findings indicate that the autoencoder learned meaningful representations that summarize participant preferences, partner evaluations, demographic characteristics, self-perceptions, and lifestyle interests.

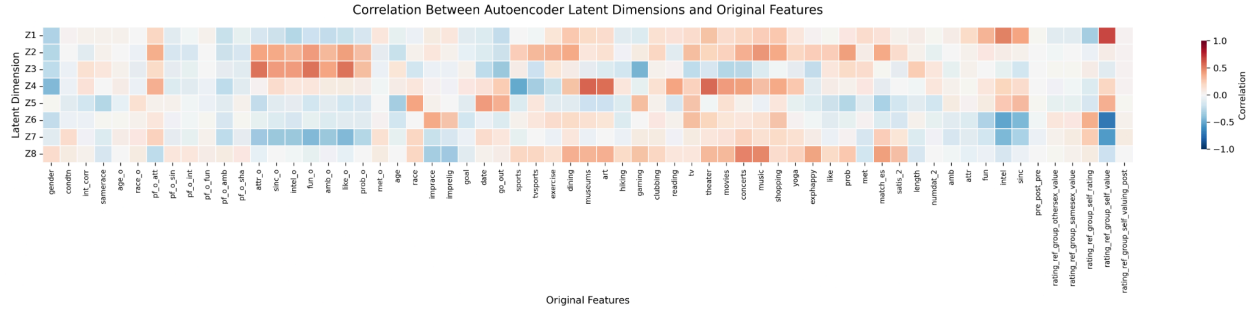


Figure 4: A heatmap where rows correspond to the eight latent dimensions Z1 - Z8 and columns to the original predictor variables (after preprocessing). Color intensity encodes the Pearson correlation coefficient, with red indicating positive and blue negative correlations.

3.3 Mixed- Effects Linear Probability Model (MELPM)

Several estimated coefficients of latent dimensions exhibited statistically significant associations with matching outcomes. Z2 showed the strongest positive association with matching probability ($\beta = 0.562, p < 0.001$), indicating that participants scoring higher on this latent dimension were substantially more likely to form successful matches. Conversely, Z4 ($\beta = -0.446, p < 0.001$), Z5 ($\beta = -0.338, p < 0.001$), Z6 ($\beta = -0.223, p < 0.001$), and Z3 ($\beta = -0.210, p < 0.001$) were associated with lower match probabilities.

The participant level random intercept variance was estimated at 0.124, which indicates meaningful between subject heterogeneity in baseline matching tendencies. This result supports the decision to use mixed-effects models rather than ordinary regression approaches that assume independent observations. Gender was also significantly associated with matching outcomes ($\beta = -0.220, p < 0.001$). Several interaction terms between gender and latent dimensions reached statistical significance, including interactions involving Z1, Z2, Z3, Z4, and Z5. These findings suggest that some latent characteristics influence matching behavior differently for male and female participants.

However, the linear probability model exhibited known limitations such as predictions occasionally falling outside the valid probability range of [0,1], requiring clipping prior to evaluation. Additionally, both AIC and BIC returned undefined values, indicating instability in likelihood based model assessment. These issues motivate comparison with the logistic mixed-effects model.

3.4 Mixed-Effects Logistic Regression (MELogR)

The Bayesian mixed-effects logistic regression produced qualitatively similar conclusions while respecting the binary nature of the outcome variable. The strongest positive latent predictor was again Z2 (posterior mean = 4.320), followed by Z3 (posterior mean = 2.621) and Z8 (posterior mean = 1.543). Strong negative effects were observed for Z4 (-2.943), Z5 (-2.891), and Z7 (-1.027). As seen when looking at the interpretations of these latent dimensions, Z2, the

strongest predictor, corresponds to immediate partner attraction and perceived romantic compatibility. The estimated participant level random effect variance parameter indicated substantial heterogeneity across individuals, further supporting the hierarchical modeling framework.

Gender continued to exhibit a strong negative association with matching probability, while several gender by latent feature interactions remained sizable. The strongest interaction occurred for gender (-4.055), suggesting that the influence of the latent characteristics captured by Z3 differs considerably across genders. Unlike the linear probability model, the logistic model produced valid probabilities directly and yielded stable likelihood-based predictions across all observations.

3.5 Predictive Performance

Model performance was evaluated on previously unseen participants contained in the test set.

Metric	MELPM	MELogR
AUC-ROC	0.681	0.703
Accuracy	0.795	0.808
Brier Score	0.158	0.159
Log Loss	1.650	0.595

Table 3: Predictive Performance on the Held Out Test Set.

The discriminative performance of the two models is further visualized by the receiver operating characteristic curves in **Figure 5**. The MELogR achieved an AUC of 0.703, compared to 0.681 for the MELPM, reflecting consistently superior separation of matched and unmatched interactions across classification thresholds.

The modest difference in AUC is consistent with the numeric metrics in **Table 3** and confirms that the logistic model provides better ranking of match probabilities. Similarly, classification accuracy improved from 79.5% to 80.8%. While the absolute improvement appears modest, it is notable given the substantial class imbalance present in the data. The logistic model also achieved dramatically lower log loss (0.595 versus 1.650), indicating better probability calibration and reduced overconfidence.

The Brier scores were nearly identical between models, suggesting that both approaches produced reasonably calibrated probability estimates overall. Nevertheless, the combination of higher AUC, higher accuracy, and substantially lower log loss indicates that the mixed-effects logistic regression provided the superior predictive framework.

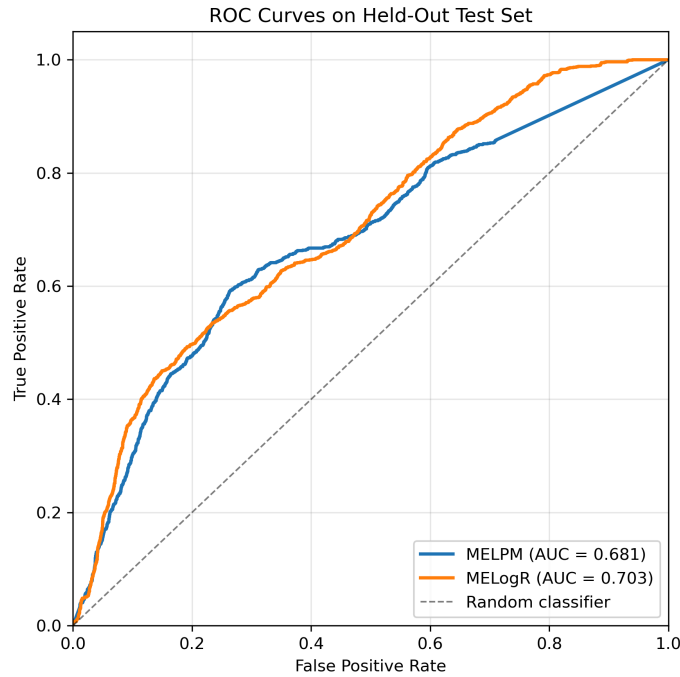


Figure 5: Overlaid ROC curves for the Mixed-Effects Linear Probability Model (MELPM) and the Bayesian Mixed-Effects Logistic Regression (MELogR) on the held-out test set, with AUC values annotated.

4. Discussion

Speed-dating is a unique social setting where individuals are thrust into a position of soul-searching, both for their own soul and for the soul meant to accompany them. It provides a unique peek into the influential structures that motivate mate selection and allows for gender differences to be examined in an experiment-like manner. One of the most interesting findings of the current study illustrates the strong impact of real-time partner evaluations on match selection, even including the post-event information.

Features reflecting real-time interpersonal reactions were more predictive than static demographic characteristics. Based upon the results, this paper has successfully proven the strong predictive abilities of a MELogR model after comparison with a MELPM with a small set of latent features collapsed and extracted by an autoencoder. This paper's investigation has led to substantial evidence supporting the idea of a multidimensional approach to predicting matches based on mate preferences, demographics, and self-perceptions. While Fisman et al. were interested in how preferences impact matching, this paper aimed to understand how these preferences are embedded in larger, psychosocial dimensions rather than acting as isolated factors. This supports the larger indication of a multidimensional approach when working with highly structured, dimensional data sets., especially in the temporal context. Temporal impact, while interesting, was not further explored here. However, results suggest future research continues this work via understanding how time impacts matching prediction.

Traditionally, social-science approaches to survey data have been restrictive and unable to capture the larger latent features at play. The latent dimensions may capture emergent effects that are difficult to measure directly through survey methodologies. While tradition sees variables selected based solely on research and conceptual relevance, the autoencoder instead allowed the current investigation to learn latent structures from the data directly. The interactivity of predictors is recovered in the current methodological approach, as demonstrated by the correlations between dimensions Z1 - Z8 representing the larger thematics captured by predictors that, previously, were treated as individual, non-conflicting factors. These dimensions capture broader concepts of interest, such as lifestyle alignment and social compatibility, that are of higher interest than singular ideas, such as “attractiveness”, that can be harder to interpret. Our finding that the strongest match predictor for the logistic regression model is the Z2 dimension, which was interpreted based on post-hoc correlations as immediate partner attraction and perceived romantic compatibility, provides an interesting insight into a specific granular dimension that is an especially effective factor in speed dating matches. Given the setting of a speed dating event, initial attraction and perceived compatibility as a strong predictor is an unsurprising, yet nuanced outcome, given the extent of possible variation in the data. By focusing on our dimensions as factors, we capture these correlations with more specificity.

The dimensions, while not explicitly visualizing unknown gender differences, rather reinforced the validity of the assumption of baseline mate selection differences between male and female identifying people even after dimensionality reduction. Future research could add further detail to these dimensional dynamics by using graph and network theory to investigate the temporary social network created by speed-dating events to isolate social structures that drive mate selection alongside latent features.

By combining unsupervised machine learning with hierarchical statistical modeling, this paper has shown that romantic selection and attraction is better understood in multidimensional processes of interconnected features. The current study is most limited by loss of and/or missing information. Much of the original data set is plagued by missingness due to a variety of factors, including survey attrition, introduction of questions throughout the event, etc. By reducing features via relevance selection and autoencoder collapsing, information is inevitably lost. The collapsing of dimensions brought upon the revelation of hidden interactions otherwise hard to capture, and simultaneously, has reduced the complexity of the data’s story by simplifying interactions into larger categories trying to proxy the same complexity. Future research should pursue the ability to preserve data complexity while advancing the dimensional features of the autoencoding process.

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